

Aggregate miRNA pressure — definitions, tests, results

Reference card. Minimal prose. Chronological design log: AGGREGATE_FORCE_VS_ABUNDANCE_DESIGN.md.

Code: gene_force_concentration.py, genome_wide_promiscuity.py. Outputs: output/coupling_predictor_compar

1. Cohorts

cohort	source	n	use
tumor	TCGA-BRCA primary, Normal-like excluded	per-gene partial-corr n=1077 (common miRNA∩mRNA∩covariate set; MIN_N=40)	ρ_tumor
healthy	GTEX breast	346 miRNA, 514 mRNA, 327 paired (SML- RNA∩RNASEQ)	ρ_GTEX, healthy baseline h

2. Expression units & normalization

matrix	unit
miRNA tumor	log2(RPM+1), TCGA GDC
miRNA GTEX	log2(TPM+1), joined to TCGA arm names (MIMAT-primary)
mRNA tumor	log2 normalized (D.load_rna)
mRNA GTEX	log2(TPM+1) (gene_tpm_v10_breast.gct.gz)
abundance a	log2RPM clipped at 0 (max(log2RPM, 0))
healthy baseline h	gtex_qn_baseline = per-arm GTEX-breast median quantile-normalized to the TCGA scale (healthy_anchor.py); resolves cross-platform offset; used in acquired transform

3. Evidence definitions (miRTarBase)

- **Canonical evidence_score** (config PRESSURE_EVIDENCE_SCORER = "confidence_logclass",
scorer score_confidence_logclass + ENCORI S1 boost). Formula:

$$\text{evidence_score}(m,g) = \sum_c w_c \cdot \log_{1p}(nf_c) + 0.3 \cdot \sum_c w_c \cdot \log_{1p}(nw_c) + 0.5 \cdot \text{enc_depth}(m,g)$$

$c \in \{\text{reporter, protein, rna, perturbation, binding}\}$

$w = \{\text{reporter } 3.0, \text{protein } 2.5, \text{rna } 1.5, \text{perturbation } 1.5, \text{binding } 0.5\}$ (assay directness)
 $\text{nf_c} / \text{nw_c} = \# \text{ Functional-MTI} / \# \text{ "Functional-MTI (Weak)" studies of class c}$ (weak $\times 0.3$)
 $\text{enc_depth} = \text{ENCORI CLIP read-depth for (m,g)}; \alpha = \text{PRESSURE_ENCORI_ALPHA} = 0.5$

Range on analysis edges: [2.08, 15.01]. **Justification (the run):** $\log_{1p}$ *within class* so a pair with 10 reporter studies is not $10\times$ a pair with 1 $\rightarrow$ defuses publication bias (the main reason fame, not biology, picks within-gene winners); CLIP/binding weighted 0.5 (physical proximity $\neq$ repression); weak-curator studies $\times 0.3$. Validated by the **curation-bias robustness run (MH-18, robustness_checks.py)**: the hub ranking is *not* a study-count artifact — the top miRNA ranks #1 under evidence, binary/degree, **and** TargetScan context++ (rank-corr $\rho=0.65$ evidence-vs-TargetScan). (*This replaces the build_edges default $3 \cdot \text{reporter} + 3 \cdot \text{binding} + 2 \cdot \text{protein} + 1 \cdot \text{rna} + 1 \cdot \text{perturbation}$ study-count scorer.*)

- **Analysis edge set** = miRTarBase Hallmark edges with **confidence_logclass score ≥ 2 (PRESSURE_MIN_EVIDENCE) + ENCORI boost** — what load_mirtar_edges produces. (This is an evidence-score floor, *not* the older “Functional-MTI + low-throughput” HE mask.) **5,219 edges / 1,424 genes / 789 arms; 5,101 / 1,408** after restricting to expressible arms (= the analysis set).
- weight currencies $w_{\text{eff}}(m,g)$: **validated** = $\log_{1p}(\text{evidence_score})$ (canonical, as above); **targetscan** = $\text{abs}(\text{ctx})$ where **ctx = TargetScan7 weighted context++ score** (hsa, summed over sites; more negative ctx = stronger predicted repression, so we use absolute value as a positive weight). TargetScan coverage is limited: its prediction file contains only **321 distinct hsa arms** (it scores conserved miRNA families, not all $\sim 2,600$ arms) $\rightarrow$ of the 789 analysis edge-arms, only **247 arms / 782 genes / 2,222 edges** carry a ctx score. This is why all ts_ constructions are coverage-limited.
- The genome-wide budget $D(m)$ (§4) uses the **same** canonical construction ($\text{confidence_logclass} \geq 2$ – ENCORI), so the budget currency matches the edge w_{eff} currency (fixed 2026-06-29; conclusions unchanged).

4. Promiscuity definitions

- **$D(m)$ = genome-wide binding BUDGET** = Σ over arm m ’s *entire* targetome of the per-edge weight (evidence-MASS, **not** a target count). Canonical: $\text{genome_wide_promiscuity.genome_wide_budget}$.
 - validated: $\Sigma_{\{\text{genome-wide, confidence_logclass} \geq 2 + \text{ENCORI}\}} \log_{1p}(\text{evidence})$ — 1,036 arms, median $D=3.55$, top = **miR-21-5p** (study-biased)
 - targetscan: $\Sigma_{\{\text{genome-wide}\}} \text{abs}(\text{ctx})$ — 320 arms, median $D=70$, top = **miR-15/16/195** (seed-targetome)
- **budget-split weight** $w(m,g) = w_{\text{eff}}(m,g) / D(m)^{\alpha}; \alpha \in \{0, 0.5, 1\}$. $\alpha=0$ = raw weight (no promiscuity); $\alpha=1$ = fraction of budget.

5. Predictor constructions (gene aggregate $P(g,s) = \Sigma_{\{m \in R(g)\}} c_m(s)$)

name	$c_m(s)$	axis
base	a	abundance dose
$\text{val_a0} / \text{val_a1}$	$a \cdot \log_{1p}(\text{ev}) / a \cdot \log_{1p}(\text{ev}) / D_{\text{val}}$	weight / weight+promiscuity
$\text{ts_a0} / \text{ts_a1}$	$a \cdot \text{abs}(\text{ctx}) / a \cdot \text{abs}(\text{ctx}) / D_{\text{ts}}$	weight / weight+promiscuity

name	c _m (s)	axis
acq	max(a - h, 0)	acquired (supra-healthy) magnitude
acq _{ev} / acq _{ts}	max(a-h,0) · log1p(ev) / max(a-h,0) · abs(ctx)	acquired × weight

6. N-effective abundance concentration (descriptor, not a test)

- **Computed on cohort-MEDIAN abundance per arm** (a = median_s log2RPM), i.e. **one value per gene from the median-abundance contribution mass — NOT a per-sample mean of n_{eff}**.
- per-gene contribution shares $p_m = c_m / \sum c$ (with $c_m = \text{median-abundance} \times W$); **Hill spectrum**: $H_0 = n_{\text{regulators}}$, $H_1 = \exp(-\sum p \cdot \ln p)$, **$H_2 = 1 / \sum p^2$ (primary)**, $H_\infty = 1 / \text{top1_share}$.
- **Purpose**: moderator / falsification. $n_{\text{eff}} \approx 1 \Rightarrow \text{aggregate} \approx \text{one edge} \Rightarrow \text{edge lemma} \Rightarrow \text{construction cannot beat abundance}$; n_{eff} high $\Rightarrow \text{construction can act}$. Tells you WHERE a win is possible.

construction (multi-reg genes)	H1	H2	H ∞	frac H2 $\geq$ 2
base log	2.81	2.53	2.04	60%
base raw-linear	1.64	1.37	1.19	21%
force _{val} log	2.79	2.45	1.90	59%
force _{ts} log	1.87	1.66	1.36	27%

Budget-split reorders top regulator vs abundance: validated 33%, ts 27%. **Use log abundance** (raw-linear collapses to single-driver). Single-driver genes: force $\equiv$ base by Spearman scale-invariance.

7. Estimator & confounds

- coupling ρ = **partial Spearman(predictor, gene mRNA | confounds)**; more negative = stronger repression. MIN_N=40.
- **canonical tumor confounds = CPE + HRD + sequencing-batch + target-gene-CN + proliferation** (proliferation = E2F+G2M metagene; canonical here, the load-bearing confound per MH-78). The level-coupling head-to-head (Test 1) uses this full set.
- **$\Delta\rho$ confounds = “symmetric composition + proliferation”**, defined precisely:
 - *composition* = epithelial + immune + stromal **marker metagenes** + proliferation metagene. **Source**: hand-curated marker gene sets (EPI_MARKERS / IMMUNE_MARKERS / STROMA_MARKERS in cross_state_coupling.py; proliferation = E2F+G2M), each a mean-z over its markers from the cohort’s mRNA.
 - *symmetric* = the **same** signatures computed **identically from each cohort’s own mRNA**, so ρ_{tumor} and ρ_{GTEx} are adjusted by an identically-constructed frame (required for the difference to be comparable). Tumor-only CPE/HRD/target-CN/batch are **excluded** (no GTEx analog).
 - *Why not CIBERSORTx?* CIBERSORTx fractions exist **for TCGA only** (annotations/cibersortx_brca_parti a prior core-coupling CIBERSORTx deconv retest is at output/core_coupling_deconv_retest_cibersortx/) They are a more principled composition estimate and **should be used as a sensitivity for the tumor-only Test 1**, but the $\Delta\rho$ needs *symmetric* covariates — using CIBERSORTx would require running it on **GTEx** with the same signature matrix (not done). The metagene was chosen because it is trivially symmetric across cohorts.

8. Controls

set	definition	n (scored)
positive	curated breast oncomiR-repressed TSGs, literature-derived (non-circular); annotations/breast_oncomir_tsg_positive_set.tsv	18
negative	miRNA-poor genes ($n_{\text{regulators}} \leq 1$)	648
orthogonal positive	gene_roles TSGs (malignancy_sign=-1)	~44

TEST 1 — level coupling ρ_{tumor} (full head-to-head; canonical confounds incl. proliferation)

- **Statistic:** ρ_{tumor} = partial Spearman(P, mRNA | CPE+HRD+batch+target-CN+proliferation). Three reads:
(a)

net-repressed = $\rho < 0$ AND BH-FDR $q < 0.05$ (FDR per predictor on the partial ρ); (b) positive-vs-negative

Mann-Whitney (unpaired, across gene sets);

- (b) **paired per-gene test:** for each gene scored under *both* predictor and base, take the difference $d_g = \rho_{\text{pred}}(g) - \rho_{\text{base}}(g)$; **Wilcoxon signed-rank** on $\{d_g\}$ (alternative $\text{pred} < \text{base}$, i.e. predictor more repressive); **bootstrap** = resample genes 2,000 $\times$ $\rightarrow$ 95% percentile CI of median d_g ; * = CI entirely below 0 $\Rightarrow$ predictor robustly beats base *gene-by-gene*. (Paired $\Rightarrow$ more powerful than (b); run on positives, the orthogonal TSG set, and all genes.) (*Supersedes the earlier no-proliferation run; proliferation is canonical here.*)

construction	gw net-FDR	med ρ pos	med ρ neg	pos<neg MWU	paired vs base: pos / TSG / all
base	444	-0.161	-0.004	1.1e-05	—
val_a0	444	-0.178	-0.004	6.5e-06	$p=0.051$ / ns / $\Delta \approx 0 \rightarrow$ inert
val_a1	412	-0.081	-0.004	6.6e-04	+0.038 ($p=0.99$) $\rightarrow$ harmful
ts_a0	304	-0.163	-0.005	3.5e-04	$p=0.64$ / ns / ns $\rightarrow$ inert
ts_a1	295	-0.088	-0.005	6.2e-03	worse $\rightarrow$ harmful
acq	460	-0.182	-0.002	5.0e-06	8.2e-4* / 4.7e-4* / 1.3e-14*
acq_ev	459	-0.193	-0.002	3.7e-06	7e-4* / 2e-4* / 2.9e-13*
acq_ts	309	-0.185	+0.001	1.1e-04	0.24 / 0.089 / 0.041 $\rightarrow$ not robust

paired median $\Delta\rho$ (acq-base): positives -0.018 [-0.026,-0.007]; TSG -0.004 [-0.023,-0.0001]; all -0.0001 (tiny, ultra-consistent).

- **Verdict:** weighting (evidence & TargetScan) **inert**; promiscuity **harmful**; **acq / acq_ev the only robust wins**; $\text{acq_ev} \approx \text{acq} \Rightarrow$ the acquired transform is the lever, weight adds nothing.

- **Limitation:** sees only *modulated* repression (needs cross-sample variance), **blind to constitutive repression** (low-variance-high-mean acquired arms invisible — burden axis untested); low sensitivity (miRNA <2-fold vs transcriptional variance); mRNA not protein; positive n=18 (orthogonal TSG + genome-wide back it); ts coverage 320 arms/777 genes.

TEST 2 — $\Delta\rho = \rho_{\text{tumor}} - \rho_{\text{GTEx}}$ (the change)

- **Statistic:** per-gene $\Delta\rho$; **NO per-gene FDR on $\Delta\rho$** . Significance = **Mann-Whitney** (positive $\Delta\rho <$ negative $\Delta\rho$). Confounds = symmetric composition+proliferation (§7). (Each cohort ρ has its own partial p; the difference is tested at the gene-set distribution level.)

predictor	med ρ_{tum} (pos)	med ρ_{GTEx} (pos)	med $\Delta\rho$ (pos)	med $\Delta\rho$ (neg)	pos<neg MWU
base	−0.097	−0.016	−0.082	−0.007	0.0028
force_val_a0	−0.102	−0.017	−0.086	−0.007	0.0021
acq	−0.124	−0.028	−0.114	−0.011	0.0022

- Validated positive control fires (positives separate from miRNA-poor, $p \approx 0.002$ – 0.003); force ties base; acq biggest effect.
- **Limitation:** GTEx-side $\max(a_{\text{gtex}} - h, 0)$ is ill-posed (rectify healthy vs own baseline → near-degenerate) so $\Delta\rho$ double-counts the anchor for acq → **ρ_{tumor} (Test 1) is the cleaner metric for acq**; no per-gene FDR; GTEx n smaller (less power on ρ_{GTEx}); constitutive-repression blindness inherited.

Verdict

acq = $\sum \max(a - h, 0)$ is the one construction that beats summed abundance (robust, proliferation-survived, modest). Edge weighting and promiscuity do not. **Open gap:** coupling cannot see constitutive (low-variance) acquired pressure — the **burden** axis (mean acquired pressure, no correlation) is not yet tested against the controls.